
FANVERSE: Exploring and Extending Branching Story Universes Across Canon and Fan Fiction

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Abstract

Fan-written works and their shared canonical source collectively form a story universe of alternative narrative branches, yet existing platforms present them as separate linear stories, leaving readers to mentally reconstruct this universe across works. We thus introduce FANVERSE, an AI-powered interactive system that transforms these works into an explicit branching story universe for integrated reading and authoring. FANVERSE organizes each branch into branch-time states linked to supporting passages and aligns branches through shared events and divergence relations. Readers can move across branches to examine how narrative changes propagate through characters, relationships, knowledge, and events, and create a grounded new branch from any selected state. Technical evaluations show that the FANVERSE pipeline supports reliable story universe construction, improved cross-branch reasoning, and more consistent branch extension. An exploratory case study further suggests that connecting interpretation with creation supports fandom sensemaking, with implications for designing fan-work engagement experiences that give users greater agency to navigate and shape story universes.

1 Introduction

Fan fiction is a major form of participatory culture that turns engagement with a source work into an open-ended, collective exploration of its story universe. Millions of fans participate in this exploration as readers, authors, or both: readers follow narrative branches imagined by others, while authors create branches not yet taken. Together, these branches form a story universe of parallel narrative possibilities. Making sense of this story universe is fundamentally relational: each branch gains meaning from what it preserves or revises relative to the source and to other branches [Derecho, 2006, Samutina, 2016]. Tracing these relationships as a reader, and extending them as an author, lies at the heart of fan-fiction engagement.

Yet existing systems support fandom engagement with individual works far better than exploration of the story universe formed by relationships among their branches. Platforms such as AO3 help readers discover related works through extensive metadata and tagging systems [Fiesler et al., 2016, Johnson, 2014, Silberstein-Bamford, 2024], but still present the works themselves as separate linear texts. Readers must therefore reconstruct shared histories and trace how a divergence ripples through everything that follows. Recent AI-assisted fan-fiction authoring systems reproduce this limitation: they can generate the next passage, but offer little support for situating a new branch within the broader story universe—for deciding what to preserve, what to change, and how those changes should propagate [Yuan et al., 2022, Chung et al., 2022, Mirowski et al., 2023, Ran et al., 2025]. Whether reading or authoring, fans therefore lack a representation of the story universe that connects the source work and fan-fiction works as branches they can compare and extend.

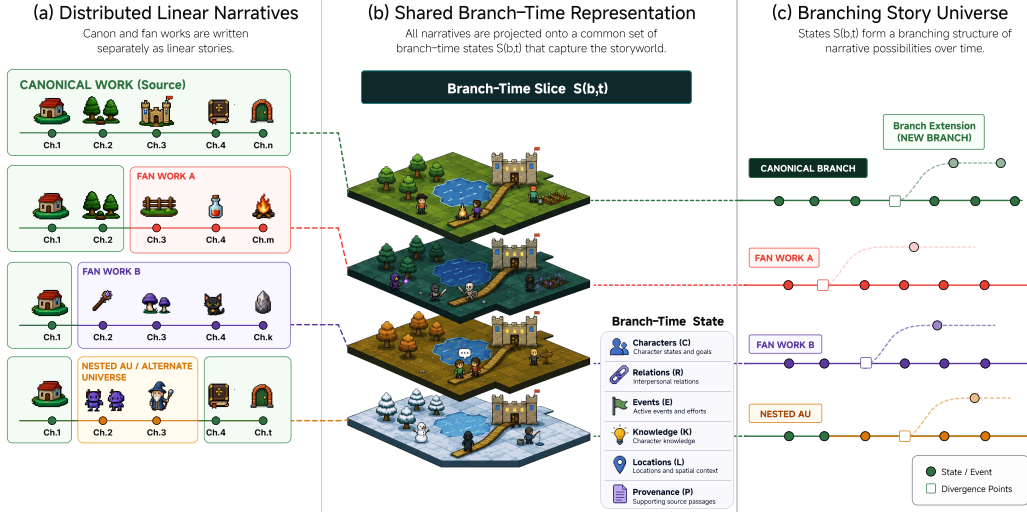


Figure 1: Overview of FANVERSE. The system transforms canonical and fan-written works into a branch- and time-specific story universe that exposes shared histories, divergence points, and branch-time states with source provenance. Readers can explore relationships among existing branches and extend a selected state into a new branch.

In response, we present FANVERSE, an LLM-powered system that transforms canonical and fan-written works into a branching story universe that readers can explore and extend. Rather than treating each work as a separate linear story, FANVERSE connects related branches through shared events and divergence points, allowing readers to follow how different possibilities unfold from a shared history. As readers move across branches, they can return to the corresponding source passages and, from any selected narrative state, extend the story into a new branch grounded in the history it inherits. In this way, FANVERSE connects the exploration of existing branches with the creation of new possibilities within the same story universe.

To support these interactions, FANVERSE transforms separate linear stories into a shared narrative graph in three steps AI-assisted pipeline. It first segments each branch into *narrative slices*, with each slice represented by a branch-time state. It then connects these slices according to story time within each branch, before aligning related slices across branches through shared anchors, inheritance, and divergence relations. Together, these relations organize how each branch unfolds and how different branches connect, while each branch-time state retains its supporting source passages. The resulting representation allows readers to trace shared histories and divergent trajectories across branches and provides the grounding for branch extension.

Our work contributes: (1) FANVERSE, an interactive system that enables users to explore and extend the story universe formed by canonical and fan-written works; (2) an AI-assisted pipeline for constructing branch- and time-specific story-universe representations that preserve shared histories, divergent states, and source provenance; and (3) technical and exploratory evaluations showing the feasibility of the representation for story-universe construction, cross-branch reasoning, and grounded branch extension.

2 Related Work

2.1 Fan Fiction as an Archive of Divergent Story Universes

Fan studies describe fan fiction as an open archive in which new works extend and transform a shared story universe [Derecho, 2006, Samutina, 2016]. Platform research examines how archives such as AO3 support participation and discovery through community-designed infrastructure and tags [Fiesler et al., 2016, Johnson, 2014, Silberstein-Bamford, 2024]. Computational work further shows how narrative attention shifts across canon and fan fiction, and provides tools for character-centric extraction [Milli and Bamman, 2016, Yoder et al., 2021]. Together, these studies portray fandom as an evolving network in which shared characters and events take on multiple trajectories. FANVERSE builds on this view by making those trajectories available for cross-work exploration.

68 2.2 AI-Supported Narrative Exploration and Understanding

69 Narrative visualizations externalize changing character, temporal, and spatial relations [Liu et al.,
70 2013, Kim et al., 2017, Hulstein et al., 2022]. More recent systems use language models to extract and
71 explore narrative fragments, knowledge graphs, and multi-level literary trajectories [Ammanabrolu
72 et al., 2020, Ye et al., 2025, Yeh et al., 2026]. These systems demonstrate how structured representa-
73 tions can support close reading and navigation. FANVERSE extends this approach to relations among
74 canonical and fan-written works, where a shared event can lead to distinct branch-specific states.

75 2.3 AI-Assisted Creative Writing

76 AI writing tools support author-controlled continuation and revision through conversational editing,
77 visual trajectory control, or hierarchical planning [Yuan et al., 2022, Chung et al., 2022, Mirowski
78 et al., 2023]. BookWorld grounds generation in an established novel by simulating a fictional
79 world [Ran et al., 2025]. Together, these systems show how structured context can guide creative
80 development. FANVERSE brings this support into fan-fiction authoring: readers compare existing
81 branches, select a branch-time state, and then create a traceable new branch from that state.

82 3 Challenges of Experiencing Fan Work as a Story Universe

83 Fan works allow readers to experience alternative narrative branches within the possibility space of a
84 canonical work. By moving across works, readers compare how possibilities unfold and interpret
85 their connections through shared characters, events, and plots. However, existing fan-fiction platforms
86 mainly rely on recommendations and tagging, leaving readers to reconstruct this story universe across
87 scattered linear works. Moreover, readers and authors often overlap in fan communities: authors read
88 others’ fan works, while readers may also create branches of their own. As AI-assisted authoring
89 becomes more accessible, creating new narrative branches has become another way to explore the
90 story universe. Unlike unconstrained story generation, such creation is grounded in the story universe
91 formed by canonical and fan-written works and in how a new idea relates to existing branches.

92 These observations motivate us to design an integrated story universe experience for fan-fiction
93 consumers with the following design goals:

94 **DG1: Faithful Story Universe construction.** We define the story universe as a unified structure
95 that faithfully captures each canonical and fan-written work as a narrative branch, together with their
96 shared elements and relationships. The system should construct this representation from the original
97 linear works without losing their narrative content or connections.

98 **DG2: Cross-branch exploration.** Readers experience the story universe by moving across narrative
99 branches and comparing how different possibilities unfold. The system should enable readers to
100 navigate, compare, and interpret branches and their relationships.

101 **DG3: Grounded branch extension authoring.** Authoring a new narrative branch within the story
102 universe is a situated process shaped by existing works and their relationships. The system should
103 embed authoring within the existing story universe, allowing authors to extend a selected branch
104 while examining its history and related branches.

105 4 FANVERSE: System Design and Implementation

106 4.1 System Design via Walk Through

107 We illustrate FANVERSE through Zoe, a *Harry Potter* fan-fiction reader who is particularly interested
108 in the Harry–Hermione pairing and wants to explore how their relationship develops across different
109 fan-written branches, then creating an alternative ending of her own.

110 As shown in Figure 2, Zoe begins with *Story Universe Selection* (a), where she enters an existing
111 *Harry Potter* story universe. If the story universe has not yet been created, she can instead use *Story*
112 *Universe Construction* (b) to construct one from the canonical work and related fan works.

113 After entering the story universe, Zoe first arrives at the *Atlas* (c). Wanting to revisit the original story,
114 she selects the canonical work and clicks *Read* to open the *Reader Page* (d), where she browses its
115 chapters and begins reading as usual. Alongside the text, the *State Inspector* (f) shows the characters,

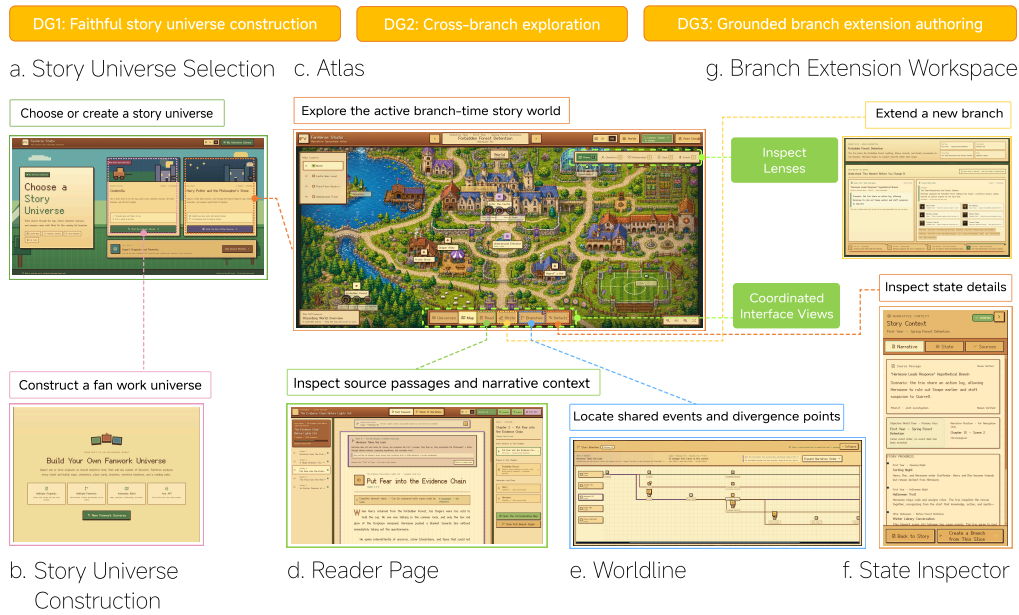


Figure 2: Overview of the FANVERSE interface organized around the three design goals. For DG1, readers can select an existing story universe (a) or construct one from canonical and fan-written works (b). For DG2, the Atlas provides the central spatial view of the active branch and story time (c), while the Reader exposes the associated source passage and narrative context (d), the Worldline aligns shared events and divergence points across branches (e), and the State Inspector presents structured state details and provenance (f). For DG3, the Branch Extension Workspace allows readers to extend the selected branch-time state into a new child branch (g).

relationships, and other state information associated with the current moment. As she reads, Zoe becomes particularly interested in how Harry and Hermione’s relationship develops and returns to the *Atlas* to explore it within the larger story universe.

In the *Atlas*, Zoe sees the characters, locations, and events around the current story moment. She notices the *Halloween Troll Incident*, a key event involving both Harry and Hermione, and selects it to return to the corresponding passage in the *Reader*. Unsatisfied with how the canonical story develops from this point, she wonders whether other fan works imagine a different continuation.

Zoe therefore opens the *Worldline* (e) to look for other alternate-universe (AU) branches that develop this moment differently. She finds the Teacher AU, a fan-written branch that imagines an alternative version of the story, where Harry and Ron seek help from Professor McGonagall (P4) instead of warning Hermione directly, and continues reading along this trajectory. By moving between the aligned states and their source passages, Zoe traces how the two choices lead to different later relationships and events. This comparison helps her understand why the canonical story develops as it does, but she is not fully satisfied with the Teacher AU either. She begins to wonder how she would continue the story herself.

Zoe then opens the *Branch Extension Workspace* (g) from the state she is viewing. Before making a change, she can review the current branch-time state and its supporting source passage. She then describes what she wants to change in natural language, while specifying how closely the new branch should follow the existing story. Based on this intervention, FANVERSE proposes corresponding state changes and subsequent events. Before writing, Zoe reviews and edits the proposed changes and events, including any conflicts or uncertainties. Once satisfied, she confirms the plan and FANVERSE generates an editable continuation, which she can revise before adding it to the story universe as a new branch.

In this way, Zoe moves from reading and comparing existing branches to developing a new possibility directly from the story state she is exploring.

4.2 Technical Architecture

This section describes our technical pipeline supporting the features described above, focusing on making explicit the fan story universe distributed across narrative branches by transforming separate linear stories into a shared narrative graph via the following three stages:

Structuralization. Given a collection of branches, FANVERSE first segments each branch into a sequence of *narrative slices*—a small unit of narrative within bounded time. For each slice, FANVERSE instantiates the branch-time state $S(b, t)$ introduced above. Formally, $S(b, t)$ is a structured vector containing six components: (1) locations, (2) character states and goals, (3) interpersonal relations, (4) character knowledge, (5) active events and their effects, and (6) supporting source passages. For example, around the *Halloween Troll Incident*, one slice captures the moment when Harry and Ron realize that Hermione is unaware of the troll but have not yet decided how to help her. At this point, $S(b, t)$ captures the shared state before their choice, which later develops into different branches: in one, they warn Hermione directly; in another, they seek help from Professor McGonagall. After structuralization, FANVERSE creates a set of narrative slices connected via the discourse order E_d of each branch.

Within-Branch Temporal Ordering. FANVERSE then connects narrative slices to reflect the chronology of each branch. For example, a childhood event appearing late in a branch as a flashback will be linked to when it actually happens. FANVERSE thus assigns story-time relations between narrative slices: {before, same-time, after, unknown}. These relations form the story-time edge set E_w and are constructed separately within each branch.

Cross-Branch Relation Alignment. Finally, FANVERSE aligns narrative slices across branches. We define a *story anchor* as a recognizable event or branch-time state shared across multiple branches. Anchors typically come from the canonical work, but shared events that emerge across multiple fan-written branches can also serve as anchors; branch-specific events after divergence do not. This design follows narrative accounts of pivotal events as points from which alternative plot trajectories can emerge. FANVERSE represents three types of cross-branch relations. {shared-anchor, inherits-from, diverges-from}. For example, two branches may reach the same decision point where the protagonist needs to make a choice, and this decision point is the shared-anchor of the two branches. These cross-branch relations form the edge set E_x .

The resulting representation is a graph $G = (V, E_d, E_w, E_x)$, where V contains the narrative slices, E_d preserves discourse order, E_w represents within-branch story-time relations, and E_x represents cross-branch relations. The edge types support complementary ways of navigating the story universe: E_d follows each work as originally presented, E_w follows events in fictional chronology, and E_x reveals shared histories and divergent trajectories across branches.

5 Evaluation

5.1 Technical Evaluation: How Well Does the Technical Architecture Support FANVERSE

We conducted three technical evaluations to examine how effectively our pipeline supports the three system design goals: story universe reconstruction (DG1), cross-branch reasoning (DG2), and grounded narrative branch extension (DG3). Details of the datasets, task construction, evaluation settings, and metrics are provided in App. A.

Table 1: Evaluation outcomes across four LLM backbones: (a) narrative-object construction, (b) cross-branch reasoning in exact accuracy (EA) and evidence F_1 (EF1), and (c) branch extension in valid-output rate (VR), overall quality, and state-error rate (SE); $\uparrow$ / $\downarrow$ shows higher/lower is better.

(a)		(b)			(c)			
Narrative object	F1	Condition	EA $\uparrow$	EF1 $\uparrow$	Condition	VR $\uparrow$	Overall $\uparrow$	SE $\downarrow$
Character	99.7%	Global RAG	79.2%	89.5%	Intent Only	82.8%	4.220	30.7%
Location	98.7%				Raw Context	92.7%	4.682	15.1%
Event	100.0%	BT RAG	82.5%	94.6%	BT RAG	94.3%	4.612	12.5%
Relationship	97.5%	BT State + RAG	80.8%	94.2%	Summary + RAG	89.6%	4.643	14.6%
Knowledge	98.1%				BT State + RAG	94.8%	4.683	6.8%

181 Tab. 1a shows near-perfect recovery of the core narrative-object catalogs (F_1 from 97.5% to 100.0%),
 182 indicating that FANVERSE can reliably recover the core narrative objects needed to construct the
 183 story universe. In Tab. 1b, Branch-Time RAG improves exact accuracy and evidence matching
 184 over Global RAG while using fewer retrieved passages. Adding structured $S(b, t)$ further improves
 185 fact F_1 , but does not improve exact accuracy over Branch-Time RAG, suggesting that branch- and
 186 time-specific retrieval provides the primary benefit for cross-branch reasoning. Finally, Tab. 1c shows
 187 that structured $S(b, t)$ most clearly improves branch-time consistency during grounded narrative
 188 branch extension. Compared with the baselines, Branch-Time State + RAG reduces state errors while
 189 maintaining a high valid-output rate.

190 5.2 Exploratory Case Study: How Do Participants Use FANVERSE

191 We conducted two approximately 45-minute exploratory sessions as the case study, based on the
 192 *Harry Potter* corpus and two participants who are familiar with both the canonical story and fan
 193 works. In each session, participants were instructed to locate a key event, compare the Canon with
 194 multiple alternative branches, to freely follow a character or event of interest, and to extend a selected
 195 narrative moment into a new branch. We present preliminary findings from these sessions to illustrate
 196 how participants used our system to explore the fan story universe through an integrated reading and
 197 authoring experience.

198 **Branch exploration revealed overlooked story details.** Cross-branch exploration helped participants
 199 trace and understand how a local change shaped later relationships and events. At the Halloween troll
 200 incident, P2 described the Teacher AU as “*a safer choice*” so chose this branch, where Harry and
 201 Ron seek help from Professor McGonagall, but noted that this choice made the three characters take
 202 “*longer to establish trust*.” This allowed participants to connect an early divergence with consequences
 203 that unfolded much later in the story.

204 **Alternative branches reframed existing characters.** Cross-branch exploration also led participants
 205 to reinterpret familiar characters, not just compare different plot outcomes. In another branch, where
 206 the students come to trust Snape earlier, P1 noted that he “*might actually be protecting them*” and
 207 could later join their confrontation with Quirrell. Encountering this expanded role, P2 reacted: “*Did*
 208 *Snape have this much going on earlier?*”. The alternative trajectory made visible how changes in
 209 knowledge and trust could recast the same character from a suspected threat into a possible protector
 210 and ally.

211 **Fandom sensemaking through authoring.** After relationships became legible, both participants
 212 gravitated toward authoring new possibilities. P1 created a branch by revealing information earlier
 213 in the Canon storyline and judged its outcome “*within a reasonable range*” because it gave the
 214 characters “*more opportunities to communicate*.” P2 likewise identified a moment where they “*might*
 215 *write*” and selected a “*completely divergent*” continuation. In this way, authoring served as a form of
 216 fandom sensemaking: readers could probe how a familiar storyworld might unfold differently.

217 6 Conclusion and Future Work

218 We presented FANVERSE, an AI-assisted system for tracing and extending branching story universes
 219 across canonical and fan-written works. By organizing related narratives through branch-time
 220 states, FANVERSE connects shared histories, divergence points, branch-specific states, and source
 221 provenance within a common representation.

222 Our evaluations show that this representation supports reliable story universe construction, improves
 223 cross-branch reasoning when combined with branch-time retrieval, and reduces contradictions during
 224 grounded branch extension. Our case studies further show how readers used these connections to
 225 trace how local changes propagated through later relationships, events, and character interpretations.

226 Together, these results suggest that reading, comparison, and creation can operate over the same
 227 evolving story universe. Although our evaluations are limited to a small set of story universes and
 228 exploratory participants, we will invite more people to experience more complete universe. Future
 229 systems could let this universe grow with fandom by incorporating new branches created from existing
 230 $S(b, t)$. They could also make exploration more experiential, allowing readers to move through the
 231 story universe much like navigating a game world.

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Table 2: Overview of the datasets used in the three technical evaluations.

Evaluation	Dataset composition	Evaluation targets
Story universe construction	A bilingual corpus containing one English canonical work and three Chinese fan-written works, including one nested fan branch; 18 annotated scenes, 35 character/location entities, and 185 atomic targets.	Narrative-object recovery, branch topology, provenance, and dynamic branch-time fields.
Grounded branch extension	Six storyworlds, 12 branch snapshots, and 24 one-sentence extension intents. Twenty intents come from synthetic unseen storyworlds and four from <i>Harry Potter</i> ; 12 target first-level branches and 12 target nested branches.	Output validity, state errors, branch-time consistency, marker coverage, and overall generation quality.
Cross-branch reasoning	Four fully synthetic storyworlds with 30 questions spanning 12 reasoning categories: current state, negative knowledge, causality, temporal order, branch comparison, knowledge state, unanswerability, state change, precondition, epistemic conflict, divergence, and provenance.	Exact accuracy, fact F_1 , evidence F_1 , evidence availability, and character-knowledge accuracy.

A Supplementary Evaluation Details

This appendix provides additional details of the datasets, task construction, evaluation settings, metrics, and judging procedures for the three technical evaluations reported in Section 5. These evaluations examine story universe construction (DG1), cross-branch reasoning (DG2), and grounded narrative branch extension (DG3).

A.1 Technical Evaluation Setup

A.1.1 Evaluation Datasets

We construct a separate evaluation set for each technical evaluation because the three tasks require different forms of narrative evidence and supervision. Table 2 summarizes their composition.

A.1.2 Common Model Settings

Across the controlled evaluations, we use four language models: `deepseek-v4-flash`, `kimi-k2.6`, `qwen3.7-plus`, and `glm-5.2`. Thinking or reasoning modes are disabled, provider-default sampling parameters are used, and the output schema is fixed within each experiment. Within an experiment, all conditions share the same model, task, output schema, and decoding settings and differ only in the context being ablated. Results reported in the main paper are aggregated across the four models.

A.1.3 Story Universe Construction

To evaluate DG1, we run the complete production construction pipeline rather than an evaluation-specific extractor. Starting from the raw narrative texts, each model independently performs narrative chunking, structured extraction, work-level consolidation, cross-work temporal alignment, and final story-universe assembly.

We evaluate construction at two levels. First, we measure recovery of the narrative-object catalogs for characters, locations, events, interpersonal relationships, and character knowledge using precision, recall, and F_1 . These metrics examine whether the pipeline recovers the core narrative objects required by the branch-time representation.

Second, we evaluate more fine-grained structural and dynamic fields, including parent-child branch topology, event discourse mode, relationship-strength bands, provenance chapters, knowledge holders, participant locations, fan-to-canon temporal relations, and canonical-event attachments. These diagnostics distinguish successful object recovery from the more difficult problem of assigning dynamic story information to the correct branch-time state.

Each constructed output is independently audited by two judge models that did not generate that output. The judges receive the relevant source passages, a finite gold checklist, and the constructed output. Predictions not included in the gold checklist are not treated as errors when they are supported by the source text. Agreement between the two judges is 98.3% for gold-target coverage and 97.5% for source support.

350 The primary catalog F_1 results are reported in Table 1a. Detailed catalog metrics and dynamic-field
351 diagnostics are reported in Sections A.2 and A.3, respectively.

352 A.1.4 Cross-Branch Reasoning

353 To evaluate DG2, we construct a controlled branch-time reasoning benchmark designed to go beyond
354 direct fact lookup. The benchmark contains 30 questions across 12 reasoning categories: current state,
355 negative knowledge, causality, temporal order, branch comparison, knowledge state, unanswerability,
356 state change, precondition, epistemic conflict, divergence, and provenance.

357 We compare three retrieval and representation conditions:

- 358 • **Global RAG**: the model receives the top five passages retrieved without restricting narrative
359 branch or story time.
- 360 • **Branch-Time RAG**: retrieval is restricted to passages belonging to the target branch and
361 valid at or before the target story time.
- 362 • **Branch-Time State + RAG**: the model receives the same branch-time-specific retrieved
363 evidence together with the structured branch-time state $S(b, t)$.

364 The Branch-Time RAG and Branch-Time State + RAG conditions use the same branch-time-specific
365 evidence and the same average number of retrieved passages. With four models answering all 30
366 questions, each condition contains 120 model-question pairs.

367 We report exact accuracy, fact F_1 , evidence F_1 , evidence availability, and character-knowledge
368 accuracy. We additionally inspect difficult reasoning categories and check generated answers for
369 unsupported facts, future information leakage, and cross-branch contamination. Detailed results are
370 reported in Section A.4.

371 A.1.5 Grounded Narrative Branch Extension

372 To evaluate DG3, we construct a controlled one-sentence-to-prose generation benchmark. For each
373 of the 24 branch-extension intents, each of the four models generates two independent outputs under
374 five grounding conditions, yielding

$$4 \times 24 \times 5 \times 2 = 960$$

375 candidates in total.

376 The five conditions are:

- 377 • **Intent Only**: the model receives only the one-sentence extension intent and the default
378 generation settings.
- 379 • **Raw Context**: the model receives the extension intent together with unfiltered narrative
380 context around the selected moment.
- 381 • **Branch-Time RAG**: the model receives the extension intent together with source passages
382 restricted to the target branch and story time.
- 383 • **State Summary + RAG**: the model receives the same retrieved passages together with a
384 natural-language summary constructed from the current state facts.
- 385 • **Branch-Time State + RAG**: the model receives the same retrieved passages and the same
386 state facts represented in the structured $S(b, t)$ format.

387 The last three conditions receive identical retrieved passage IDs. The last two conditions additionally
388 receive identical atomic fact IDs. This design controls for retrieval quantity and fact content when
389 comparing alternative state representations.

390 All 960 generated candidates are retained under an intention-to-treat protocol. Invalid, incomplete, or
391 unparseable outputs are counted as failures rather than removed or regenerated.

392 Each candidate receives a blinded primary judgment from a model different from the generator.
393 The evaluation rubric contains five 1–5 ratings: intent completion, branch-time consistency, context
394 use, causal coherence, and narrative quality. Judges additionally annotate explicit error categories,

Table 3: Detailed accuracy of narrative-object construction.

Narrative object	N	Precision	Recall	F1
Character catalog	144	99.4%	100.0%	99.7%
Location catalog	136	98.1%	99.3%	98.7%
Event catalog	144	100.0%	100.0%	100.0%
Relationship catalog	72	96.4%	98.6%	97.5%
Knowledge catalog	80	100.0%	96.3%	98.1%

Table 4: Fine-grained structural and dynamic diagnostics for story-universe construction.

Target	Correct / Total	Accuracy
Parent-child branch topology	12 / 12	100.0%
Event discourse mode	129 / 144	89.6%
Relationship-strength band	63 / 70	90.0%
Provenance chapter	96 / 144	66.7%
Knowledge holder	65 / 153	42.5%
Participant location	116 / 336	34.5%
Fan before/same/after temporal relation	31 / 96	32.3%
Canonical-event attachment	13 / 96	13.5%

including wrong branch, future leakage, knowledge leakage, wrong location, relationship conflict, unsupported facts, and missing authorized changes.

We use valid-output rate and judged state-error rate as the primary outcomes. Overall quality, branch-time consistency, and marker coverage are treated as diagnostic measures. For paired comparisons, we additionally report case-clustered bootstrap 95% confidence intervals. Detailed diagnostic results are reported in Section A.5.

A.2 Construction Metrics

Table 3 provides the precision, recall, and sample counts underlying the narrative-object reconstruction results reported in Table 1a. Here, N denotes the number of evaluated instances for the corresponding narrative-object category.

A.3 Construction Diagnostics

High catalog-recovery scores do not imply that all dynamic fields are aligned equally well. Table 4 therefore reports additional atomic and topological diagnostics.

These results distinguish catalog recovery from dynamic alignment. The production pipeline reliably recovers narrative-object catalogs and parent-child branch topology, and performs relatively well on event discourse mode and relationship-strength bands. In contrast, assigning participants to the correct event-level locations, identifying knowledge holders, and aligning fan-written events with canonical temporal anchors remain substantially more challenging.

A.4 Cross-Branch Reasoning Breakdown

Table 5 provides the detailed results of the controlled cross-branch reasoning evaluation reported in Table 1b.

Branch-time filtering reduces the mean retrieved context from 5.00 to 3.13 passages while increasing evidence availability from 92.7% to 100.0%, evidence F_1 from 89.5% to 94.6%, and exact accuracy from 79.2% to 82.5%. Adding the structured branch-time state further increases fact F_1 from 98.1% to 98.8%, but does not improve exact accuracy or character-knowledge accuracy over Branch-Time RAG.

The benchmark also contains deliberately difficult reasoning categories. Exact accuracy on pre-condition questions is 16.7% for Branch-Time RAG and 25.0% for Branch-Time State + RAG; on epistemic-conflict questions, the corresponding accuracies are 62.5% and 37.5%. No condition

Table 5: Diagnostic breakdown of controlled branch-time reasoning.

Condition	N	Mean passages	Evidence available	Exact accuracy	Fact F1	Evidence F1	Knowledge accuracy
Global RAG	120	5.00	92.7%	79.2%	98.1%	89.5%	56.8%
Branch-Time RAG	120	3.13	100.0%	82.5%	98.1%	94.6%	65.9%
Branch-Time State + RAG	120	3.13	100.0%	80.8%	98.8%	94.2%	59.1%

Table 6: Diagnostic outcomes for controlled branch extension.

Condition	Branch-time consistency	Marker coverage
Intent Only	4.245	83.8%
Raw Context	4.714	87.5%
Branch-Time RAG	4.682	87.8%
State Summary + RAG	4.677	89.8%
Branch-Time State + RAG	4.729	88.1%

424 produces unsupported selected facts, future leakage, or cross-branch contamination in this controlled
425 evaluation.

426 A.5 Branch-Extension Diagnostics

427 Table 6 reports diagnostic outcomes that complement the valid-output and state-error results in
428 Table 1c. Branch-time consistency is the rubric score for alignment with the selected branch-time
429 state. Marker coverage measures whether the target state markers specified for an evaluation instance
430 are reflected in the generated continuation.

431 Compared with Intent Only, Raw Context increases the overall quality score by 0.463 (95% CI [0.285,
432 0.621]) and reduces judged errors by 15.6 percentage points (95% CI [-20.8, -9.9]), showing that a
433 one-sentence intervention alone provides insufficient grounding.

434 With retrieved evidence controlled, Branch-Time State + RAG does not significantly improve the
435 overall score over Branch-Time RAG (+0.072, 95% CI [-0.054, 0.193]). Its primary benefit is instead
436 consistency: the judged error rate decreases from 12.5% to 6.8% (a difference of -5.7 percentage
437 points, 95% CI [-9.9, -1.6]). Compared with State Summary + RAG, which contains the same atomic
438 facts, the structured state decreases the judged error rate from 14.6% to 6.8% and increases the
439 valid-output rate from 89.6% to 94.8%.

440 The trend is also observed outside well-known fiction. Across the 20 synthetic unseen cases, the
441 judged error rate decreases from 28.8% under Intent Only to 6.3% under Branch-Time State + RAG;
442 across the four known-fiction cases, it decreases from 40.6% to 9.4%. For first-level branches, the
443 corresponding decrease is from 25.0% to 6.3%, and for nested branches from 36.5% to 7.3%.

444 A.6 Judging Protocol and Reliability

445 For story universe construction, two independent judge models that did not generate the evaluated
446 output perform a source-constrained semantic audit. Each judge receives the relevant source passages,
447 a finite gold checklist, and the constructed output. Predictions absent from the gold checklist are
448 retained as correct when supported by the source text. The two judges achieve 98.3% raw agreement
449 on gold-target coverage and 97.5% agreement on whether additional predictions are supported by the
450 source.

451 For branch extension, every candidate is anonymized and receives a primary judgment from a model
452 that did not generate it. All 960 candidates are retained when computing valid-output and state-error
453 rates. An additional 178 candidates receive an independent second judgment. The double-judged
454 overall scores have Pearson correlation $r = .40$, and exact agreement on the complete set of error
455 labels is 54.5%.

456 Following prior work on LLM-based evaluation of open-ended outputs, we use the second judgment
457 to audit potential position, verbosity, and self-enhancement biases Zheng et al. [2023]. Given the
458 moderate agreement on subjective judgments, we treat rubric scores as diagnostic rather than as

459 ground truth and emphasize structured validity and error outcomes in the main conclusions Artstein
460 and Poesio [2008].

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